



Rationally or emotionally: how should voice user interfaces reply to users of different genders considering user experience?

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Received: 9 March 2021 / Accepted: 10 September 2021 / Published online: 27 September 2021
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Abstract

Voice user interfaces (VUIs) have exploded in popularity over the past 3 years. However, there has been little research on the reply methods that VUIs can adopt to communicate with people. In this paper, we designed 2 studies with 20 participants to explore the influence of reply methods on user experience in 2 kinds of scenarios (applicational scenarios and giving a command) when using a VUI. We explored the performance of different reply methods (fact-only, rational, and emotional) at different times and in different scenarios. In addition, we examined whether there were gender differences when evaluating a reply and different preferences for different reply methods. A “Wizard of Oz” method was used in the experiments to simulate real scenarios for communication between the participants and the VUI. We divided a reply into three parts (fact + judgment + strategy) and constructed three kinds of reply methods. In the experiments, we used quantitative scoring (five aspects: affection, confidence, naturalness, social distance, and satisfaction), preference selection and an interview to measure the participants’ user experience. The results indicated that the participants were inclined to prefer the reply methods (rational and emotional) that offered judgments and strategies in our experiment script, and the emotional style received the highest evaluation. In addition, we found that male participants tended to have a higher evaluation of VUIs’ replies for all three reply methods in applicational scenarios and when giving a command than female participants in our studies. In general, these results may contribute to the design of VUI replies.

Keywords Voice user interface · Smart speaker · Reply methods · User experience

1 Introduction

With the continuous development of voice interaction technology, people tend to use existing intelligent assistants as effective life assistants. In recent years, commercial voice user interface (VUI) products with intelligent voice assistants (VAs) have become available in the consumer market.

Various personal voice assistants continue to appear, such as Google Home, Siri (America) and Xiaomi, Xiaodu (China). Users commonly like to listen to music, search for information, and automate home Internet-of-Things (IoT) devices, with reported usage rates of 40%, 17%, and 9%, respectively (Bentley et al. 2018). In recent studies, VUI has also been used for education (Melton and Fenwick Jr 2019), monitoring mental health (Maharjan et al. 2019), and reducing public speaking anxiety (Wang et al. 2020). However, with the continuous improvement of VUIs, users not only require that products have the characteristics of usability and ease of use, but also hope that products can provide a satisfactory affective user experience (UX) (Park et al. 2020).

There are many different factors that can influence user experience with VUIs. With regard to activation, different wake words and personal pronouns (Yanyan et al. 2019), as well as different activation methods (Biermann et al. 2019) and feedback time (Coskun-Setirek and Mardikyan 2017), can influence the user experience with VUIs. In terms of voice dialogue, different response time (Lee et al. 2020) and

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language expression factors, such as speaking speed (Yanan et al. 2019) and filler words (Jeong et al. 2019), are also factors that can have an influence on user experience. Moreover, the progressivity principle and speech design (Fischer et al. 2019) can affect the user experience, and the virtual character of VUIs (Jeong et al. 2019; McKeown 2016) can also affect the user experience. Moreover, integrating relevant strategies of human communication into VUIs can improve user experience (Cohen et al. 2004). Therefore, integrating the communication style into the VUI may also affect the user experience. However, it is still rare to integrate reply methods associated with different communication styles into the human–computer interaction of VUIs.

Therefore, we constructed three kinds of reply methods for the design of different communication styles. Two studies were designed to compare performance in different scenarios. These reply methods were labeled the fact-only, rational, and emotional styles. We divided a reply into three parts: fact, judgment, and strategy. The fact-only style just stated a fact, while rational and emotional styles were composed of fact + judgment + strategy. The emotional style was more emotional, and the rational style was more formal. In addition, we explored gender differences in the user experience for each reply.

2 Related work

2.1 Conversational agents and communication style

The Computers Are Social Actors (CASA) paradigm suggests that human–human and human–computer interactions share similar characteristics. In the process of human–computer interaction, the same social rules and expectations that humans would apply to humans will also be applied to computers (Cohen et al. 2004). In our previous studies and expert discussions, we realized that as conversation agents, VUIs could have their own communication style, which may be a variable that affects users' experiences with VUIs.

Several studies have revealed that human-like communication strategies are critical because they help people build social attributions with agents, resulting in more trust and more frequent system use (Jang 2020; Kopp et al. 2005). In addition, the progressivity principle (Fischer et al. 2019) and different transparencies (Karsenty and Botherel 2005) can help the VUI designer improve their products. Pearl (2016) also proposed and outlined principles for designing VUIs, such as setting user expectations, confirmation and disambiguation.

Since human-like strategies could help VUIs improve their user experience, we explored the influence on user experience of the agents having a communication style

similar to that of human beings. There are different dimensions in different communication theories. Norton (1978) distinguished nine communicator styles, and Merrill and Reid (1981) developed the social style model (SSM) based on two fundamental dimensions of social behavior: assertiveness and responsiveness. According to the SSM model, the first dimension represents how dominantly and self-confidently a person communicates, and the latter dimension focuses on emotional reactions to communication. We can also say that people with different responsiveness have different characteristics of rationality or more thinking and emotion.

2.2 User's gender and VUI user experience

Users of different genders may have different experiences with the same VUI or reply because male and female users may have different requirements and feelings for VUIs. Some studies about human–computer interaction have proven that users' gender has an effect on interface preferences (Schwind et al. 2017) and how artifacts are designed (Schwind and Henze 2018). Recently, some findings have provided evidence that male users had higher willingness than female users in terms of VUI use and VUI use continuance (Nguyen et al. 2019). Besides, some researchers showed that male users rated VUI's usability higher than female users (Habler et al. 2019). The gender effect of user experience may be related to the gender of the agent's voice: female participants accepted female robots more easily than male robots, and male participants accepted male robots marginally more than female robots. And male users' ratings of female voices were lower than female users' ratings (Eyssel et al. 2012). To date, there is still a paucity of work examining the gender effects of VUI in the literature, so examining the role of gender on attitude towards VUI use is a meaningful research topic.

Previous studies have mainly focused on the principles and strategies of designing VUIs. However, it is still rare to combine communication style with VUI replies. Our study constructed three kinds of reply methods based on different communication styles and compared the user experience with each in an experimental manner. In addition, we explored the role of gender on attitude towards VUI replies, which is meaningful for designing VUIs for users of different genders.

3 Research questions

The current study assessed the effects of the reply methods associated with different communication styles and genders in different scenarios on UX. Specifically, we focused on the following two questions.

1. Do different reply methods have a main effect on user experience?
2. Do users of different genders have different user experience trends and preferences for different reply methods?

4 Prestudy: the usability test

Before our research, we tested the usability of four typical smart speakers (Xiaodu at Home, Xiaodu, Xiaoi, and Tmall Genie) from the perspective of quantitative evaluation and collected user evaluation data on system usability, voice quality, and affective experience. The experimental materials for this experiment were all in Chinese. In the usability test, we tested and analyzed seven kinds of problems (i.e., activation methods, recognition, feedback, appearance, sound, privacy, and screen problems) existing in speakers and analyzed the severity of these problems.

4.1 Method

4.1.1 Participants

In total, 38 participants took part in this usability test. The proportions of student participants and office workers were balanced. Since the usability test was for smart speaker users, we tended to recruit participants who had experience using smart speakers. Most participants had experience using smart speakers.

4.1.2 Experimental methods

This usability test mainly aimed to evaluate four different speakers through quantitative questionnaire evaluations and qualitative interviews. In this usability test, we used three existing scales [i.e., System-Usability-Scale (SUS, Brooke 1986), MOS-Expanded (MOS-X, Polkosky and Lewis 2003), Pleasure-Arousal-Dominance (PAD, Mehrabian 1996)] that are frequently used to collect participants' ratings of speakers when they use different functions.

The SUS scale was compiled by Brooke in 1986 and consists of ten questions. This scale was used in our study to measure the system usability of smart speakers. The MOS-X scale was compiled by Polkosky and Lewis in 2003 to measure the quality of synthesized sounds. The MOS-X scale is composed of 15 questions and is divided into 4 dimensions.

There are different ways to evaluate emotions. The Reverse Comic Strip (Vanderhaegen 2021), SELF-ASSESSMENT-MANIKIN (Bradley and Lang 1994) method and other methods can be used to evaluate emotion. The PAD scale can be used to evaluate the emotions inspired by pictures (Miccoli et al. 2016) and sounds (Krause and North

2017). As this study mainly studied the reply of smart speakers, we chose PAD as our tool to measure emotion.

The simplified PAD emotion scale is a scale for measuring emotion developed by Mehrabian based on the PAD three-dimensional emotion model (three dimensions: pleasure, arousal, dominance). Since the dimensions of arousal and dominance do not intuitively reflect the user's experience, the PAD scale scoring results mainly focus on the pleasure score.

4.1.3 Experimental design

Since the user's previous use experiences with a smart speaker needed to be considered, each participant needed to test a speaker that he or she had never used before. We recorded the speakers that the participants had used, and selected speakers that they had never used before. In addition, each participant tested only one speaker; thus, the usability test adopted a between-subjects design.

Dependent variables Since different questionnaires were used in this test, we were able to obtain quantitative data about system availability, voice quality, and emotional experience. In addition, from the semistructured interviews, we were able to discern the problems that the participants focused on when using smart speakers and the severity of those problems.

Procedure Before the test, the participants filled out the informed consent form and the confidentiality commitment form, and basic information was introduced. After that, they filled out the basic user information questionnaire and participated in a short interview before the test. After the experiment officially started, the participants were asked to interact on-site in different scenarios. After all the experience tasks were completed, they were asked to fill out the three questionnaires. The process lasted 150 min for each participant.

4.2 Results

According to the ANOVA results of the SUS scale data, there was no significant difference between the SUS scores of the four speakers, and Xiaodu obtained the highest score; the MOS-X scale data showed that all the speakers performed better in terms of intelligibility and performed poorly in terms of naturalness, rhythm and social impression. Among the four smart speakers, the Xiaomi speaker performed the worst in all aspects; in terms of intelligibility, naturalness, and rhythm, Xiaodu at Home performed best. In all dimensions, the performance of Xiaodu speakers and Tmall Genie was nearly the same; according to the ANOVA results of the PAD scale data, the user experiences with the four speakers were all pleasant, and Xiaodu received the highest evaluation.

The results also indicated that there were many problems in replies when the participants communicated with the speaker. Then, we further counted the frequencies of different problems. Among them, repetitive and inflexible replies (23.7%), inability to adjust for user groups (26.3%), inappropriate recommendation scenarios and content (15.8%), and too simple answers (21.1%) were the main reasons that reduced the user experience of the participants. In addition, in the interviews, the participants reported that the lack of emotion in the speaker's answer made them feel uncomfortable.

4.3 Discussion: design of the reply methods

The realization reply method was determined by the expert group through brainstorming after the test by the expert group. After several rounds of brainstorming, the expert group had a full discussion and finally reached agreement. The smart speaker's reply could be composed of "facts + judgments + strategies". The speakers first state the facts (such as the title of the song), then add their judgments on the current situation, and then give their suggestions. The speaker's reply can be disassembled into three parts. The first part is a fact, which is the realization of a basic function, such as "check the weather" and "play music"; the second part is a judgment based on the facts of the current situation, such as "it will rain tomorrow" and "it is very late now"; the last part is a strategy, which involves proposing solutions to current problems after making a judgment, such as "take care of yourself; don't get sun-stroke!" and "come on and have a rest".

In accordance with the results of the literature review, for this study, we constructed operational definitions for "emotional style" and "rational style" as shown in Table 1.

4.4 Discussion: determination of experimental scenarios and scenario time

In the usability test, the results also indicated that the participants tended to use functions such as weather, music, news, and alarm clock, which were regarded as

applicational scenarios. The common times that users used VUIs were the morning, the afternoon (after work), and at night before bedtime. In addition, we found that there were some functions that were often used, such as adjusting the speech rate. There were two kinds of instructions for users with regard to adjusting the speech rate. We defined them as "precise instructions" and "vague instructions", respectively. Their definitions were as follows: the precise instructions outlined to users generally how to adjust the speaker's speech rate by describing precise numbers, for example, "adjust the current speed to 1.5 times" and "adjust the speed to 60"; the vague instructions explained to users generally how to adjust the speaker's speed by describing the speed adjustment direction, for example, "speak faster".

5 Study 1: examination of the influence of reply method on the UX of VUIs

5.1 Method

5.1.1 Participants

Twenty participants (ten males and ten females) participated in both study one and study two. All of them were in the age range of 20–40 years. We tended to recruit participants who had experience using smart speakers and all participants recruited in this study had experience using an intelligent VUI. The proportion of college student participants was the same as that of working participants. The entire experiment lasted approximately 30 min, and each participant was compensated 30 RMB after finishing it. If the participants felt uncomfortable during the experiment, they could choose to withdraw from it at any time.

This survey was reviewed and approved by the Human Research Ethics Committee in the School of Economics and Management at Beihang University and was conducted with explicit consent obtained from each participant.

Table 1 Composition of the content of different reply methods

Reply methods	Composition of reply contents		
	Fact	Judgment	Strategy
Emotional	Statement of fact	Descriptions combined with subjective feelings	The expression is more emotional and life-like; the tone is richer, and the modal particles are used more
Rational	Statement of fact	Statement of objective judgment	The expression is formal and emotionless; the tone is stable tone and modal particles are not used or rarely used
Fact-only	Statement of fact	Not applicable	Not applicable

5.1.2 Experimental methods and equipment

A "Wizard of Oz" (WOZ) method was used in the experiments. The WOZ method involves researchers simulating the interaction process and the functions of a system "behind the scenes", causing participants to mistakenly believe that they are interacting with a real system (Bernhaupt et al. 2017; Green and Wei-Haas 1985; Kerly and Bull 2006). Taking research information recommendations as an example, a tester can set the response of the AI system according to the participants' previous choices, but the participants will think that the responses are real responses from the system. When participants think they are talking to a real system, they are able to interact more seriously and naturally. Therefore, the WOZ method is one of the most commonly used voice interaction test tools and is widely used in the fields of experimental psychology, human factor engineering, ergonomics, linguistics, and usability engineering. Besides, the sounds of the speakers used in this experiment were all the same female voice. The experimental materials for this experiment were all in Chinese.

In addition, preference selection and semistructured interviews were used to measure the user experience of the participants. After each round of experiments (after experiencing the three reply methods in a scenario and scenario time), the participants were asked to choose only one reply method they liked the most (choose one of three). In the semistructured interviews, the participants were asked the following question: which reply method did you form a deep impression about? Why?

In the experiments, a computer was used as the backstage equipment. In addition, a controllable smart speaker that could be controlled to change its test demo was used to conduct the studies. This experiment used the control platform of the Baidu AI Interaction Design Platform to control the response of the experimental equipment. The experimenter logged in to the control platform through a computer, and the smart speaker's reply, including content, volume, and speech rate, could be set in advance on the platform.

5.1.3 Experimental design

Combining the results of preliminary literature collation, usability testing, and preresearch, this study adopted an in-subject design to study the influence of reply method (fact-only style, rational style, and emotional style) on UX. The specific language design is shown in Table 2. This experiment was a within-group experiment, and all three variables were within-subject variables.

Dependent variables In this study, we measure UX from five factors: affection, confidence, naturalness, social distance, and satisfaction. These factors are derived from

related scales for evaluating speech systems or UX (Ameen et al. 2021; Hone and Graham 2000; Klein et al 2020; Radziwill and Benton 2017). Affection and confidence are important factors to evaluate the UX of users in response to the voice system (Street 1982). Naturalness can evaluate the response of the speaker system from two aspects: language expression and voice (Adiga and Prasanna 2019). Social distance is of great significance for understanding effective communication styles and coordinating interpersonal relationships (Kim and Mutlu 2014). Satisfaction is widely used as a comprehensive factor to measure UX. All factors use a seven-point Likert scale, from "1" to "7", representing "completely disagree" to "completely agree", and the specific expression of each dependent variable is shown in Table 3.

After the experiment, a Principal Component Analysis (PCA) was conducted for the five ratings, and we found that all five variables had loading weights of 0.85 or greater onto only one component, and the variance explanation rate reached 76.41% ($\alpha = 0.96$). Therefore, the average score of these five ratings was used to measure UX.

Control variables In this study, the scenarios and scenario times of this experiment were regarded as control variables. The scenarios were checking the weather and listening to music and news, and the scenario times considered were morning, afternoon and evening. All the scenarios and scenario time were introduced to the participants, not the actual usage scenarios and scenario times.

Procedure The procedure can be seen in Fig. 1. Before the experiment, a practice session that lasted approximately 5–10 min was provided for participants to become familiar with the equipment, and each participant had two opportunities to practice. Next, participants were required to complete the informed consent and a questionnaire listing basic demographic information. Then, they were introduced to how to use the smart speaker and the whole experimental process. Each time they listened to a reply from the speaker, they needed to score the reply on the five dependent factors. When the formal experiments were conducted, they completed tasks according to a fixed experimental sequence (the WOZ method and paper questionnaires were used to collect data, so the experimenter needed to set many corresponding replies in the background and perform key operations). It was complicated to adjust the order of reply methods, so the order of the speaker's reply was fixed. After each round of experiments (after experiencing the three reply methods in a scenario and scenario time), the participants were asked to choose only one reply method they liked the most. In addition, an interview was conducted to learn about participants' preferences for the reply methods and the reason for their preferences.

Table 2 Specific replies in different application scenarios and scenario times

Scenarios	Scenario time	Reply methods		
		Fact-only		Emotional
Check the weather	Morning	Description of today's weather	Description of today's weather + It's hot, pay attention to heatstroke prevention and cooling	Description of today's weather + It's so hot out there, be careful of heatstroke!
	Afternoon (after work)	Description of tomorrow's weather	Description of tomorrow's weather + you get off work, take a break	Description of tomorrow's weather + it's so late, come on and have a rest
	At night	Description of tomorrow's weather	Description of tomorrow's weather + it's twelve o'clock now, and you have to go to work tomorrow. It's time to go to sleep	Description of tomorrow's weather + it's so late, if you don't go to bed, you will be late for work tomorrow
Listen to music	Morning	Play the song— <i>Sunny Allure</i>	Good morning, hope you feel happy today (Play the song— <i>Sunny Allure</i>)	You got up so early, you must be full of energy today (Play the song— <i>Sunny Allure</i>)
	Afternoon (after work)	Play the song— <i>Summer of Kikujiro</i>	It's time to have a rest. (Play the song— <i>Summer of Kikujiro</i>)	It's so late after work today. Let's relax. (Play the song— <i>Summer of Kikujiro</i>)
	At night	Play the song— <i>Baby</i>	It's twelve o'clock, and it's time to go to bed, good night. (Play the song— <i>Baby</i>)	It's so late, you should go to bed, I wish you a good dream. (Play the song— <i>Baby</i>)
Listen to news	Morning	News content—garbage classification	You have learned about the news related to the enforcement of garbage classification. Here is the latest development for you. (News content—Garbage classification)	The garbage sorting in Shanghai has baffled many people, and now Beijing will do the same. Let's get the latest situation together. (News content—Garbage classification)
	Afternoon (after work)	News content—garbage classification	You have learned about the news related to the enforcement of garbage classification. Here is the latest development for you. (News content—Garbage classification)	The garbage sorting in Shanghai has baffled many people, and now Beijing will do the same. Let's get the latest situation together. (News content—Garbage classification)
	At night	News content—garbage classification	You have learned about the news related to the enforcement of garbage classification. Here is the latest development for you. (News content—Garbage classification)	The garbage sorting in Shanghai has baffled many people, and now Beijing will do the same. Let's get the latest situation together. (News content—Garbage classification)

5.2 Results

5.2.1 Quantitative analysis

As shown in Fig. 2, the results suggested that there were differences in UX among the different reply methods and between genders. Male participants had a higher average UX score than female participants, and the average emotional

style score was higher than the scores for the rational and fact-only styles.

To investigate the within-subject factor of different reply methods and the between-subject factor of gender, a repeated-measures analysis of variance (ANOVA) was conducted for each variable.

Reply methods and UX The static analysis indicated that the reply methods had a main effect on UX [F

Table 3 Description of variables for user experience measurement

Factor	Items
Affection	In this situation, the reply of the voice system is pleasing
Confidence	In this situation, I have confidence in the reply of the voice system
Naturalness	In this situation, the reply of the voice system is natural
Social distance	In this situation, the reply of the voice system can shorten the distance between me and the voice system
Satisfaction	In this situation, the reply of the voice system is satisfactory

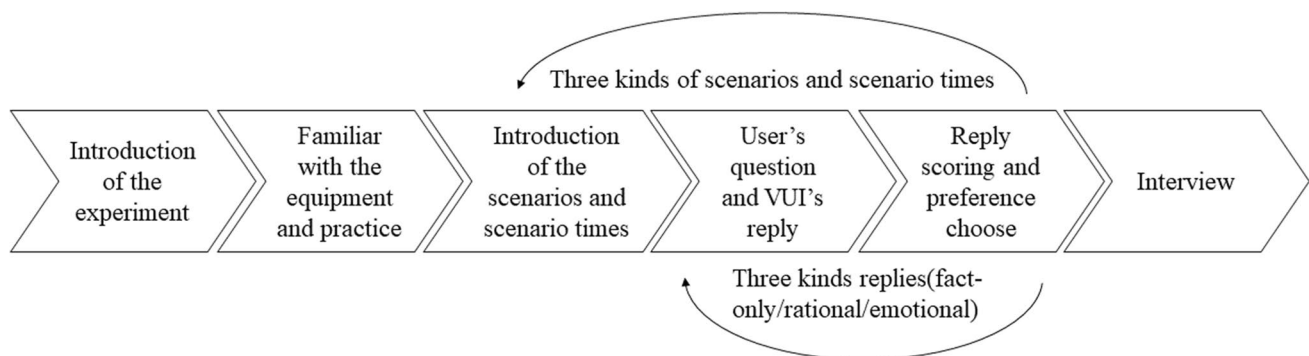
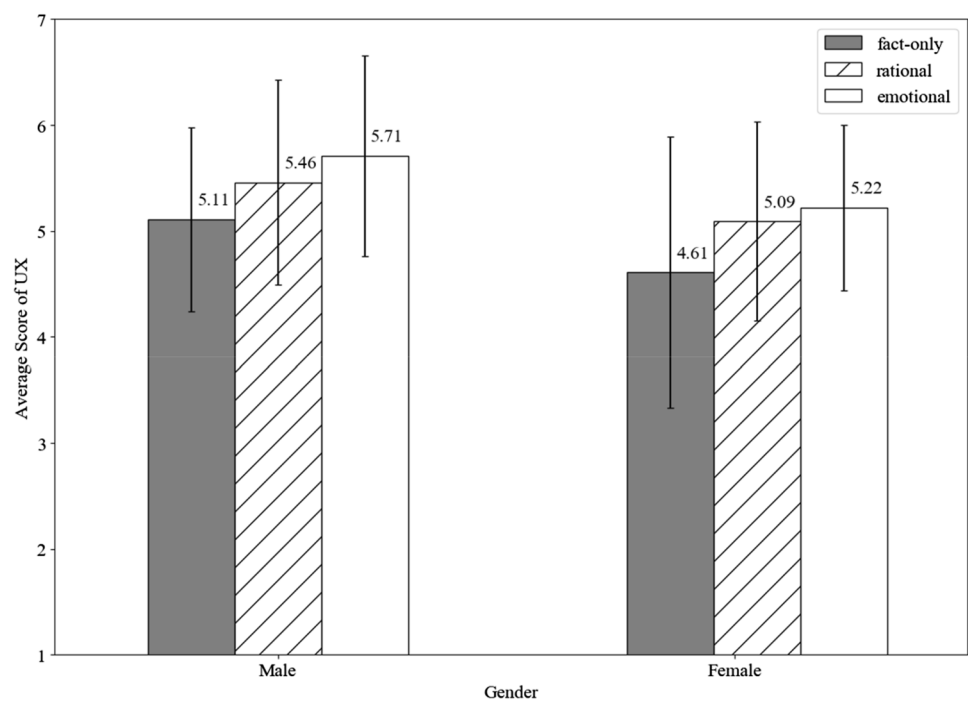


Fig. 1 The procedure for study 1

Fig. 2 Average UX score of different reply methods and genders



(2,356) = 28.481, $p < 0.001$]. The UX scores of rational and emotional styles were higher than those of fact-only styles without any further description. In addition, the UX scores of emotional style were higher than those of the other two styles [$F(1,178) = 41.230$, $p < 0.001$]. Therefore, VUIs performed better when they had a rational and emotional style than when they had a fact-only style.

Gender effect We also found that there were significant differences between the UX scores of different genders [$F(1,178) = 16.445$, $p < 0.01$]. The overall evaluation of the UX of smart speakers by male participants ($M = 5.43$, $SD = 0.96$) was significantly higher than that of female participants ($M = 4.97$, $SD = 1.06$).

Preference selection After each experiment, the participants were asked to complete the self-reported questionnaire to select their favorite reply method. Each participant was asked to choose only one reply method in a scenario and scenario time. We calculated the frequency with which each reply method was selected in all scenarios and scenario times. The detailed frequencies are shown in Table 4. In general, 18.3% of the participants chose the fact-only style, 31.1% of the participants chose the rational style and 50.6% of the participants chose the emotional style, which was consistent with the results obtained by the quantitative analysis. At different scenario times and scenarios, this trend changed. When listening to music at night, only 20% of the participants chose the rational style, while 33% chose the fact-only style. However, 35% of the participants chose the fact-only style and only 15% chose the rational style when listening to music in the afternoon. When checking the weather at night, 45% of participants chose the rational style, while 40% chose the emotional style. In other cases, the proportions were consistent with the overall trend.

5.2.2 Interview results

After each experiment, the participants were asked about their comments on the replies. The reasons for their choices and the frequencies are also presented in Table 4. In addition, we calculated the frequency of the reasons for choosing the different reply methods mentioned in the interviews. According to the results, some participants tended to choose the fact-only style (2 times), mainly because they felt that the smart speaker talked too much in these scenarios, and they felt bored or disturbed. Two participants who chose the fact-only style mentioned that they preferred the more serious speaking style. The conversation atmosphere (two times) and the limited emotional needs (four times) were also mentioned. The rational style was chosen eight times because it made the participants feel closer and more comfortable. Sometimes the rational style was chosen (three times) because the response given was relevant to the user's experience. Concision was another reason that the participants chose the rational style (two times). However, the emotional style was chosen (15 times) mainly because of its humorous and colloquial reply method, which made participants feel closer and more comfortable. The reasons and evidence for choosing different reply methods are shown in Table 4.

5.3 Discussion

Our results indicated that among the different reply methods, replies with judgments and strategies were more likely to receive higher evaluations from users than replies with only facts. Moreover, it revealed that it was far from sufficient when a VUI reply contained only basic information about the question. Users tended to prefer VUI judgments and coping strategies on the discussion topics. This trend was reflected in different scenarios and at different scenario times.

Table 4 Reasons and evidence for choosing different reply methods

Reply methods	Total frequency	Reason	Evidence
Fact-only	33	No need at night (5)	"Don't need it at night, don't like it"
		Don't want to listen to witticism (2)	"People are more energetic in the morning, and they are willing to listen to teasing, but not after work or at night"
		Fewer words (5)	"Don't like speakers that talk too much at night"
		The atmosphere was not destroyed (2)	"The song is very good, coupled with some voice intervention, it may be a kind of interruption and change the atmosphere"
Rational	56	Close and moderate (8)	"I like the second choice, it makes me feel closer and more comfortable"
		Facts and experiences related (3)	"It mentioned what I have learned before"
		Concise (2)	"Smart speakers didn't talk much and speak smoothly"
Emotional	91	Humorous and witty, colloquial (15)	"The third reply is more heart-warming, and a humorous reminder is quite interesting, bringing the distance closer"

The emotional style received the highest average UX score and was selected the greatest number of times. This result indicated that users were more inclined to have more emotional communication with VUIs (e.g., smart speakers) in their daily lives. Some researchers found that integrating emotions into conversational agents would make them more believable as social partners (Becker et al. 2007; Dybala et al. 2009). We also found that integrating emotions into reply methods can improve user experience. In our research, this kind of emotional integration was mainly carried out through the organization of language and the expression of modal particles. The VUI showed its care for users and provided friendly suggestions to them, which reduced the social distance. Therefore, it is important and effective to integrate our reply methods and emotions into VUIs.

Moreover, the interview results showed some of the reasons for the participants' choice. For most of the participants who chose the emotional style, their evaluations were mainly that this style was close and humorous, indicating that reducing interpersonal distance and enhancing the sense of humor may be why this reply method fared better than the other two reply methods. Some studies have confirmed that VUIs or conversational agents with a sense of humor are more highly rated by users (McKeown 2016; Niculescu and Banchs 2019). For those who chose the rational style, some participants thought that this strategy achieved closeness and suitable psychological distance. Compared with the other two methods, this reply method can be used on more serious occasions and in topic discussions while ensuring a better experience. Two participants also said that the reason for their choice of the rational style was that the advice provided by the VUI was relevant to their previous experience, which also reflects its effectiveness compared to that of the fact-only reply method. Although the overall user experience of the fact-only reply method was not as positive as that of the other two kinds, the reasons that the subjects chose this reply method still deserve attention. On the one hand, the results showed that the choice of this reply method may be related to scenario time and scenario. On the other hand, this choice also reflected the participants' pursuit of simplicity and their personalities. Moreover, for people with different personalities, if we could analyze their personality characteristics and combine those results with the communication styles (reply methods) of VUIs, we may achieve better results for UX.

We further analyzed some of the differences in the trends in different scenarios in this study, we still noted that there may be some differences at different scenario times and in different scenarios. From the questionnaire, we obtained the participants' reply method choices, and we found that participants expressed different tendencies towards the same reply method and tended to choose different reply methods at different scenario times and in different scenarios. In study 1, 76% of the participants chose different reply methods in

different scenarios, and 55% of the participants chose different reply methods at different scenario times. The results could be explained by changes in the needs and moods of the same person at different scenario times and in different scenarios.

Our study also explored gender effects on user experience. The results indicated that the average UX of male participants was higher than that of female participants. The gender effect in study 1 was significant, and the trend was different from the results reported by some researchers (Habler et al. 2019). Since the design of our studies mainly aimed to examine the emotion-based experience of participants in the WOZ experiment, the results differed from those obtained from studies examining the experience of answering multiple questions about the same function in only one scenario, and our results also reflected that female users have high requirements for the emotional experience of VUIs.

6 Study 2: examination of the influence of reply method on UX considering user commands

Study 1 mainly evaluated the impacts of different reply methods and genders on user experience. In study 2, the instruction types for adjusting the VUI speech rate were introduced as a new within-subject variable. The purpose of study 2 was to explore the influence of different reply methods, instruction types and genders on user experience.

6.1 Method

6.1.1 Participants, experimental methods, and equipment

The participants and the equipment for studies 1 and 2 were the same. The "WOZ" method and preference choice approach (choose the reply method the participant liked the most of three reply methods) were also used in this study. Besides, the sound of the speakers used in this experiment were all the same female voice and same to study 1. The experimental materials for this experiment were all in Chinese.

6.1.2 Experimental design

In study 2, we took adjustment of the speaker's speech rate as the main task, and the precision and vagueness of the expression of the speed were taken as between-subject variables. Therefore, a two-factor within-subject design (3×2) was used in this study to investigate the effects of the reply method (fact-only, rational, and emotional style) and instruction type (vague and precise) on UX. The specific speech designs of different instruction types are shown in

Table 5. Moreover, the participants were asked to choose the way (vague or precise) they were more accustomed to adjusting the speaker's speech rate. After each round of the experiment, the participants were also asked to choose their favorite reply style in different instructional scenarios.

Dependent variables Similar to study 1, we measured UX based on five factors: affection, confidence, naturalness, social distance, and satisfaction. Therefore, the dependent variables and questionnaire used in study 2 were the same as those in study 1. The validity of the questionnaire was verified in study 1. The average score for these five ratings was also used to measure the UX.

Procedure Because the participants in studies 1 and 2 were the same, the participants were more familiar with the experimental equipment and experimental procedures. The participants were first required to complete an informed consent and a questionnaire listing basic demographic information. The participants were also asked to choose the way (vague or precise) they were more accustomed to adjusting the speaker's speech rate. Then, they needed to ask smart speakers how to make scrambled eggs with tomatoes. After that, the speaker would reply. During this time, the participants gave instructions to adjust the speaker's speech rate. Each participant experienced two ways of adjusting the speech rate, precise and vague and three kinds of reply methods, fact-only, rational, and emotional. After each experience, the participants filled out a questionnaire to score their user experience. After each round (after experiencing the three reply methods in a scenario), they were asked to choose their favorite reply style in different instructional scenarios.

6.2 Results

To investigate the effects of the within-subject factors of instruction type and reply method and the between-subject factor of gender on the dependent variables, a repeated-measures analysis of variance (ANOVA) was conducted for each variable. All analyses were performed with IBM SPSS Statistics 25.

Instruction type and UX As described before, we classified the instructions into precise instructions and vague instructions. The main effect of instruction type on UX score was significant [$F(1,18)=9.399, p<0.01$]. In addition, the average UX score for vague instructions was significantly higher than that for precise instructions [$F(1,18)=9.399, p<0.01$], which meant that instruction type could influence users' UX. The vague speed adjustment method may be more consistent with the usual life situations, and more people are accustomed to this adjustment method.

Reply methods and UX In general, UX score increased with the order of reply method (fact-only, rational, and emotional), but the effect of communication on UX was not significant [$F(2,36)=1.649, p>0.05$]. Moreover, there was a cross effect between reply method and gender ($p=0.081$). To obtain a deeper understanding of the reason for this phenomenon, we separated participants of different genders and analyzed the trends separately.

Gender effect The same trend was observed regarding gender, namely, the average UX of male participants was higher than that of female participants, but the difference was not significant [$F(1,18)=2.552, p>0.05$]. As shown in Fig. 3, there seemed to be different trends between different genders. In the applicational scenarios, male participants felt nearly the same about all types of replies; however, female participants showed the same trend that was observed in study 1.

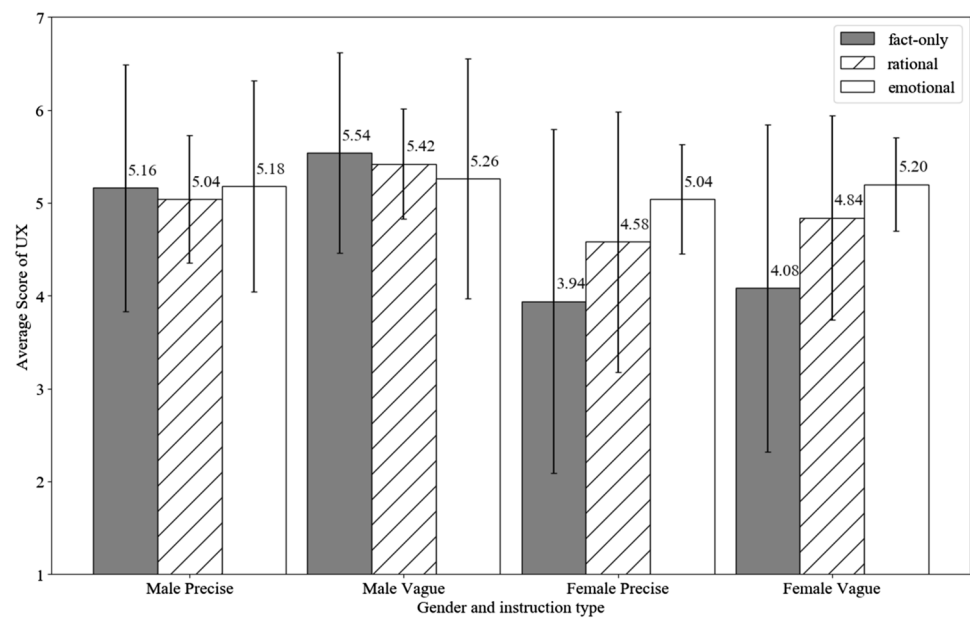
Therefore, we separated the male and female participants and conducted repeated-measures analyses of variance separately. For male participants, we found that the main effect of instruction type was significant [$F(1,9)=12.250, p<0.01$], but the UX between different reply methods was nearly the same [$F(2,18)=0.089, p>0.05$]. However, for female participants, the difference in UX between different instruction types was not significant [$F(1,9)=2.078, p>0.05$], and the difference in UX between different reply methods was just significant [$F(2,18)=3.559, p=0.05$].

Preference choice We counted the number of participants who chose different reply methods in different instructional scenarios. The number of participants who reported being accustomed to vague instructions (six males and six females)

Table 5 Specific speech design in instructional scenarios

Instructions	Reply methods		
	Fact-only	Rational	Emotional
Precise instruction (please adjust the speed to two times/adjust the speed to 50)	Content after the speed adjustment	Okay, adjusted to 1.5 times speed + content after the speed adjustment	So easy! It has been adjusted faster + content after the speed adjustment
Vague instruction (please speak faster)	Content after the speed adjustment	Okay, adjusted to 1.5 times speed + content after the speed adjustment	So easy! It has been adjusted faster + content after the speed adjustment

Fig. 3 Average UX scores of different reply methods, genders and instruction types



was higher than the number of participants who reported being accustomed to precise instructions (three males and three females). In addition, the number of participants who reported choosing the emotional style was higher than that for the other two reply methods, regardless of whether they were male or female. We also noticed that 18 participants chose the same reply method regardless of the instruction type. Therefore, we analyzed the choices of these 18 participants. The overall trend reflected in the results was mostly consistent with the UX rating data. What was inconsistent was that the male participants' UX scores for the fact-only, rational, and emotional style were nearly the same, but when they chose the reply method they preferred, they (55.6%) showed a more obvious preference for emotional style.

6.3 Discussion

In study 2, the results also indicated the same trends in some aspects: users tended to prefer replies with judgments and strategies over replies with only facts; the emotional style received the highest average UX score and was selected the most frequently; and the average UX of male participants was higher than that of female participants.

Apart from that, there were still some differences between studies 1 and 2. Our results suggested that instruction type may influence UX. In addition, in the two studies, the main effects of reply method between different genders were partly different. In study 1, both male and female participants showed significant main effects of reply method. However, in study 2, the UX difference between the reply methods of male participants was very small. This situation indicated that male participants might be not very sensitive to the different reply methods in the instruction situation, but

females had higher requirements for emotional experience and paid more attention to VUIs' ease of use. This finding is important because the requirements of VUI users of different genders may be different; therefore, we need to design for users of different genders to achieve a better user experience.

7 Discussion

Some previous studies have investigated the effects of many factors on user experience, but there have been few studies on the effect of reply method related to different communication styles on user experience. Pigliacelli (2020) constructed two kinds of conversational style (business vs pleasure style), and the former style was more serious, the latter was more emotional. However, they found that the conversational style did not have a moderating effect on either Perceived Ease of Use-Perceived Usefulness or Perceived Ease of Use-Attitude relationships. However, the impact of different communication styles on the detailed user experience has not been tested. Shamekhi et al (2016) constructed high involvement (HI) and high considerateness style and found that users exhibited similarity attraction toward computer agents, and preferred the agent whose conversational style matched their own.

In this paper, we conducted two studies to investigate the effect of different reply methods (fact-only style, rational style, and emotional style) on user experience. The communication styles were constructed differently from the above studies. The results indicated that the "fact + judgment + strategy" method could effectively improve user experience and subjective evaluations at different scenario times, in different scenarios, and with

different instruction types. Specifically, compared with the other two reply methods, the emotional style is the most effective at improving user experience.

In our daily life, VUIs, especially smart speakers, serve as life assistants. If they could communicate smoothly based on context and use the same communication styles or emotions exhibited by humans, then we could further improve the intelligence of and user experiences with VUIs. However, when there is no need for such a communication style or when most environmental information cannot be obtained, we can still use the rational and emotional reply methods.

We further analyzed some of the differences in the trends for different scenarios in the current article, we still noticed that there may be some differences for different scenario times and in different scenarios. From the questionnaire, we obtained the participants' reply method choices. Interestingly, we found that participants expressed different preferences the same reply method and tended to choose different reply methods at different scenario times and in different scenarios. In study 1, 76% of the participants chose different reply methods in different scenarios, and 55% of the participants did so at different scenario times. The results could be explained by changes in the needs and mood of the same person at different scenario times and in different scenarios. Our results also suggested that instruction type can influence UX. Combined with user preferences, we can make customizations or specific design choices based on different scenarios, scenario times, and instruction types.

We also found that users' gender may influence the user experience. On the whole, male users have higher evaluations of smart speakers' replies than female users. This trend was consistent with the trend found by Habler et al. (2019) when VUI was used for music search. Besides, this trend may also be related to the fact that male users were more willing to use smart speakers than male users mentioned by Nguyen et al. (2019). Since female voices of the smart speaker were used in our experiments, our finding was different from the findings of Eyssel et al. (2012) who found that female users had higher scores for female voices than male users. Moreover, in study 1, in daily applicational scenarios, male participants gave higher evaluations of emotional and rational replies than fact-only replies. Integrating emotions into this kind of scenario may be a good way to improve the user experience. In study 2, for instructional scenarios, male users gave almost the same evaluations to different replies, while female users still maintained their preference for emotional and rational. This showed that the preferences of male users and female users were different in different scenarios. Therefore, it may be better to consider the gender-based differentiated design of smart speakers in different scenarios when designing smart speakers.

8 Conclusions and limitations

This paper presents the impact of reply methods with different communication styles on user experience of an VUI. Our studies provide VUI designers with a new perspective regarding reply methods with different communication styles (fact-only style, rational style, and emotional style). In addition, the gender of users should be considered when designing commercial VUIs. An application of our experimental results is to combine these findings with user characteristics for further personalization and customization, which will improve the user experience. Moreover, the experiment in this article was an experiment in Chinese, and the results can be further verified in the English context to compare the results in different languages.

There were some limitations in the current study. First, the results of our studies were based on the data from 20 users. A larger sample size will be required to obtain more convincing results in the future. Second, the reply order of our experiment was fixed, further optimization is needed. Last, the wording design of our studies can be further adjusted and improved if additional research is conducted.

Funding This study was supported by the National Natural Science Foundation of China (NSFC, 72171015 and 72021001) and the Fundamental Research Funds for the Central Universities (YWF-21-BJ-J-314).

Declarations

Conflict of interest The authors declare that they have no conflicts of interest.

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